

# VocaLens — print day

Two jobs, different materials. **Start the PETG first** — it is the long one, and the TPU can run after it or on a second machine.

JOB 1 · PETG · the two pods~3 h

|                    |                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File               | FINAL_petg_plate_HEADERS.stl — your XIAO has 2×7 headers soldered on, so this is the one.<br>Print FINAL_petg_plate_XIAOSOCKETS_AMPFLAT.stl instead <b>only</b> if the amplifier has wires soldered flat to its pads rather than jumper sockets pushed onto pins. <b>If unsure, print this one</b> — a pod that is slightly too big still closes, one that is too small cannot be fixed. |
| What is on it      | 4 parts: front pod base, front pod lid, rear pod base, rear pod lid                                                                                                                                                                                                                                                                                                                      |
| Material           | PETG (PLA is acceptable if PETG is not free — it is stiffer and more brittle, but it fits)                                                                                                                                                                                                                                                                                               |
| Layer height       | 0.2 mm                                                                                                                                                                                                                                                                                                                                                                                   |
| Infill             | 25 %                                                                                                                                                                                                                                                                                                                                                                                     |
| Perimeters / walls | 3                                                                                                                                                                                                                                                                                                                                                                                        |
| Supports           | OFF                                                                                                                                                                                                                                                                                                                                                                                      |
| Brim               | ON — the pods are tall and narrow and can be knocked over                                                                                                                                                                                                                                                                                                                                |
| Plate size         | 105 × 108 × 27 mm — fits any printer at the venue                                                                                                                                                                                                                                                                                                                                        |

**Why supports off:** the plate is already arranged so nothing overhangs. The pods sit cavity-up and the lids sit flat face down. If your slicer warns about overhangs, something imported wrong — do not switch supports on to silence it.

JOB 2 · TPU · the four rings~40 min

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| File               | FINAL_tpu_rings.stl                                                                     |
| What is on it      | 4 copies of one small ring — 2 per pod, plus spares                                     |
| Material           | <b>TPU / flexible filament.</b> This one cannot be substituted                          |
| Layer height       | 0.2 mm                                                                                  |
| Infill             | 20 %                                                                                    |
| Perimeters / walls | <b>2</b> — not 3. The wall is only 1.2 mm; more perimeters make it too stiff to stretch |
| Print speed        | <b>20–25 mm/s.</b> TPU jams if pushed fast                                              |
| Retraction         | Near zero (0.5 mm or less)                                                              |
| Supports           | OFF                                                                                     |
| Plate size         | 68 × 10 × 11 mm (unchanged — same file as before)                                       |

**If the venue has no TPU,** print the rings anyway in the softest filament available and expect them to grip poorly, then say so — but ask around first. The rings are the only part that has to flex, and a rigid ring will crack rather than stretch onto the arm.

## WHEN THEY COME OFF THE BED — check in this order

1. **Slide a ring onto your glasses temple arm.** It should need a firm push and then stay put on its own.

**If it is loose:** that is the one failure that matters, and it is a 10-minute reprint. Reduce `RING_IN_VERT` and `RING_IN_OUT` in `vocalens_ring.scad` by 0.5 mm each and print one ring again. **Make the opening smaller, never larger** — this part grips by stretching, like a hair tie.

2. **Slide a pod base onto a ring's dovetail rail,** along the arm. It should need real effort and should not rattle or lift off sideways.

**If it will not go on at all,** lower `DT_SQUISH` from 0.45 to 0.35. **If it rattles,** raise it to 0.55.

3. **Drop a lid onto its base.** The rib should sit inside the cavity and the lid should sit flush.

4. **Check the front lid has two openings:** a round hole near one end for the camera, and an 18 × 20 mm oval recess near the other for the touch pad, with a small hole through the middle of the oval.

The small hole is for the touch wire. Solder the wire to the **back** of the pad *before* gluing the pad into the recess — afterwards you cannot reach it, and nothing conductive should be exposed where a finger touches.

5. **Check the rear lid** has a ring of small holes — that is the speaker grille.

## TAKE WITH YOU

- ☐ Laptop with the STL files on it (or the GitHub repo open)
- ☐ USB stick or SD card — most venue printers will not take a network job
- ☐ **The glasses you are mounting to** — the ring fit test is the whole point and cannot be done later
- ☐ Your assembled XIAO + camera, to check it drops into the front pod cavity
- ☐ The speaker and amplifier, to check they drop into the rear pod cavity
- ☐ Superglue or double-sided tape for the touch pad

**Print the TPU rings even if you are short on time.** The pods are useless without them, and they are the fastest part on the list.